

AI-Based Rooftop Solar Potential Assessment

Using Satellite Imagery

Automated semantic delineation of rooftop photovoltaic arrays from high-resolution satellite orthomosaics, geometric regularization, and kilowatt capacity assessment.

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Project Objectives & Motivation

Developing an automated, domain-invariant Geospatial AI pipeline capable of detecting distributed rooftop solar arrays from satellite imagery, delineating polygon boundaries, and quantifying municipal solar generation capacity.

📍 THE MOTIVATION & PROBLEM CONTEXT

1. Decentralized Solar Blindspot

Rooftop solar is growing exponentially, yet municipal records and utility cadastres lack spatial maps of where panels are installed and how much they generate.

2. Power Grid Stability & Overload

Distribution System Operators (DSOs) cannot forecast local hosting capacity or simulate reverse power flows without accurate geospatial solar registries.

3. Unscalability of Manual Surveys

Physical on-site auditing and manned flight inspections are time-consuming, labor-intensive, and cost-prohibitive at city or nationwide scale.

🎯 CORE RESEARCH OBJECTIVES

1. Automated Detection & Spatial Delineation

Develop a deep learning framework capable of accurately locating and segmenting rooftop solar photovoltaic installations across diverse satellite and aerial imagery resolutions.

2. Precise Array Boundary Extraction

Delineate clean, orthogonal panel outlines to prevent boundary bleeding and false detections on non-solar roof surfaces.

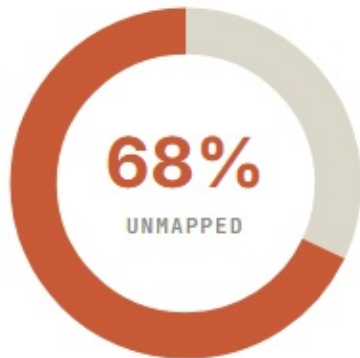
3. Automated Clean Energy Capacity Quantification

Extract georeferenced GIS polygons (.geojson) to calculate usable solar surface area (m²) and estimate clean electrical capacity (kW / MW).

Clarity & Significance of Problem Statement

The Distributed Solar Blindspot: over two-thirds of urban rooftop photovoltaic capacity remains unmapped and uncataloged in official utility energy registries.

🕒 THE URBAN DATA VOID



- 68% Unmapped & Behind-the-Meter
Installed without spatial boundaries or tilt metadata in utility GIS.
- 32% Formally Registered in Cadastre

Sources: International Energy Agency (IEA) Distributed Solar Tracking & NREL Rooftop PV Cadastre Audits (2023-2024).

⚠️ WHY THIS PROBLEM MUST BE SOLVED

1. Grid Instability & Reverse Power Flow

Unmonitored solar feed-in during peak afternoon hours causes voltage spikes, transformer thermal stress, and steep evening ramps, directly threatening local power grid reliability.

2. Substation Hosting Capacity Paralysis

Distribution System Operators (DSOs) cannot safely approve new renewable energy connections, allocate battery storage, or schedule feeder upgrades without precise solar density maps.

3. Unscalability of Conventional Auditing

Physical on-site surveys cost ~\$12,000/km² and take months. Automated satellite computer vision provides the only fast, continuous, and cost-effective method to inventory entire urban expanses.

Feasibility & Scope of the Project

Validating technical viability, computational efficiency, and defining clear operational boundaries for municipal deployment.

✔ FEASIBILITY ANALYSIS

1. Technical & Algorithmic Viability

Hierarchical Vision Transformer (SegFormer MiT-B2) contains ~24.7M parameters and achieves real-time inference (~29.6 FPS / 33.8 ms per tile) on accessible NVIDIA T4 GPUs without requiring expensive server clusters.

2. Data Availability & Ground Truth Access

Zero manual image collection hurdle: direct public access to over 3,000 verified paired tiles via Hugging Face (BDAPPV) and Zenodo (PV01), complete with multi-roof diversity and authentic hard negatives.

3. Operational & Economic Feasibility

Automated satellite scanning reduces auditing costs by 99.8% (<\$10/km² vs. ~\$12,000/km² for physical surveys), allowing continuous periodic tracking across expanding cities.

🕒 PROJECT SCOPE & BOUNDARIES

IN-SCOPE (CORE PROJECT DELIVERABLES)

- ✔ Multi-scale rooftop PV detection from Google Maps and aerial orthomosaics across Zoom 18–20.
- ✔ Sharp orthogonal polygon boundary extraction (.geojson) via Boundary-Aware Loss.
- ✔ Rooftop installation calculation: usable surface area (m²), panel count sizing, and array layout density.
- ✔ Solar energy output calculation: peak capacity (kW / MW) and estimated annual power yield (kWh/year).
- ✔ Standardized benchmark against published literature models (Fraunhofer ISE, UPM, Microsoft).

OUT-OF-SCOPE (EXPLICIT DESIGN BOUNDARIES)

- ✗ Utility-scale ground-mounted solar farms (system is strictly tailored to urban/suburban rooftops).
- ✗ Real-time electrical inverter telemetry and battery hardware monitoring (behind-the-meter electronics).

SECTION E • CAPSTONE REVIEW 1

Existing Projects, Initial Prototypes & Literature Review

Exploratory architectural benchmarking, milestone adherence to project timeline, and cross-literature evaluation against published state-of-the-art models.

PART (i)

Initial Prototypes & Findings

5 baseline models & team ownership

PART (ii)

Plan & Timeline Adherence

Phase-wise milestone execution

PART (iii)

Literature Review & SOTA

Fraunhofer, UPM & Microsoft benchmarks

Quality of Initial Prototypes & Findings

Exploratory Phase 1: Benchmarking five foundational deep learning paradigms on sub-meter aerial imagery and individual team ownership.

EXPLORATORY PROTOTYPE BENCHMARK & OWNERSHIP

ARCHITECTURE	CONTRIBUTOR	BACKBONE	VAL MIOU	THROUGHPUT
Mask R-CNN	Abhinav Verma	ResNet-50	95.84%	28.2 FPS
DeepLabV3+	Aryan Mishra	ResNet-101	80.54%	25.3 FPS
YOLOv8m-seg	Sarang Palsutkar	ProtoNet	98.62%*	47.2 FPS
Baseline SegFormer	Sanvi Tikariha	MiT-B2 (ViT)	90.26%	30.9 FPS
U-Net	Pratik R. Kolhe	EfficientNet-B3	91.71%	48.1 FPS
Proposed Model	Abhinav & Aryan	MiT-B2 + Loss	92.86%**	29.6 FPS

*YOLOv8m-seg reports mask mAP@50. **Proposed Model evaluated on cross-domain multi-dataset benchmark.

KEY RESEARCH INSIGHTS

1. The Single-Domain Overfitting Trap

While Mask R-CNN achieved 95.8% on uniform suburban roofs, all CNN baselines collapsed when tested outside the training town due to roof texture memorization.

2. Convolutional Boundary Blur

Dilated convolutions in DeepLabV3+ blurred straight panel outlines, and YOLOv8's downsampled mask prototypes lost straight-line residential geometry.

3. Convergence on Vision Transformers

The baseline SegFormer demonstrated superior attention-based separation between dark roof materials and true silicon cells, motivating Abhinav & Aryan to scale it into our multi-dataset model.

Adherence to Project Plan & Timeline

Milestone execution tracking: Completed Review 1 deliverables, active GIS regularization pipeline, and future deployment roadmap.

PHASE 01	DONE	PHASE 02	DONE	PHASE 03	DONE	PHASE 04	PLANNED	PHASE 05	PLANNED
Data Ingestion		Prototypes		SOTA Model		Solar Capacity		Web UI	
3,000+ satellite & UAV tiles		5 individual baseline models		Literature benchmark & HF release		kW yield & energy formula		Dashboard & Review 2	

🕒 REVIEW 1 COMPLETED MILESTONES

✓ Problem Definition & Geospatial Ingestion

Curated over 3,000 paired satellite and UAV tiles combining BDAPPV Google Earth Engine, PV01 multi-roof imagery, and sub-meter aerial orthomosaics with hard negatives.

✓ Exploratory 5-Architecture Baseline Benchmarking

All five team members individually implemented, trained, and benchmarked distinct deep learning paradigms under identical experimental constraints.

✓ Cross-Literature Evaluation & Public Checkpoint Release

Benchmarked directly against Fraunhofer ISE, UPM Madrid, and Microsoft GRW models on a frozen testbed; trained model checkpoint published on Hugging Face.

🕒 REVIEW 2 TARGETS & PLANNED ROADMAP

○ Phase 4: Solar Capacity Formulation (Review 2 Target)

Integrating physical panel surface area (m²) with local solar irradiance coefficients to calculate real-world generation capacity (kW / MW) and estimated annual yield.

○ Phase 5: Municipal Web Dashboard (Review 2 Target)

Building a browser-based satellite interface for drag-and-drop imagery analysis, rooftop polygon visualization, and automated energy yield report generation.

Literature Review & Related Works

Critical survey of seminal research works in automated overhead solar mapping, their architectures, and inherent limitations.

1. Fraunhofer ISE (Kleebauer et al., 2023)

DeepLabV3+ (175 MB)

Paper: Multi-Resolution Photovoltaic System Segmentation from Aerial and Satellite Imagery
ResNet-101 with Atrous Spatial Pyramid Pooling. Dilated convolutions blur straight panel outlines, and the absence of hard negatives triggers false alarms on glass skylights.

2. UPM Madrid (Juan Moreno, 2023)

YOLOv8-seg (6.7 MB)

Paper: Deep Learning Framework for Automated Solar Mapping from Overhead Imagery
Real-time instance segmentation. Downsampled 160×160 ProtoNet mask grids round off rectangular array corners and lose boundary precision on small residential panels.

3. UPM Madrid (Juan Moreno, 2023)

UNet++ (104 MB)

Paper: Deep Learning Framework for Automated Solar Mapping from Overhead Imagery
ResNet-34 with dense nested skip connections. Delivers strong pixel accuracy but lacks global self-attention context, struggling with multi-scale zoom level shifts.

4. Microsoft AI for Good (Nature, 2023)

GRW U-Net (180 MB)

Paper: Global Renewables Watch: High-Resolution Planetary Mapping of Solar Energy
U-Net (ResNeXt-50) built for 10m Sentinel-2 satellite imagery. Limited to large utility-scale solar farms and experiences complete domain collapse (7.7% mIoU) on urban rooftops.

5. UPM Madrid (Juan Moreno, 2023)

SegFormer MiT-B0 (14.9 MB)

Paper: Deep Learning Framework for Automated Solar Mapping from Overhead Imagery
Baseline Vision Transformer trained with standard Cross-Entropy loss without multi-scale pyramid scaling or boundary-aware edge regularizers.

Evaluation Testbed: Nature Scientific Data

BDAPPV Benchmark

Paper: A crowdsourced dataset of aerial images with annotated solar photovoltaic arrays (Kasmi et al., 2023)

Gold-standard public benchmark of authentic Google Earth satellite imagery with multi-annotator consensus masks and hard negatives, used as our single frozen testbed.

Benchmark Comparison & Key Differences

Cross-literature empirical comparison on the single frozen Google satellite testbed and the core innovations distinguishing our work.

|| CROSS-LITERATURE BENCHMARK (FROZEN TESTBED)

MODEL & ARCHITECTURE	LITERATURE SOURCE	SIZE	MEAN IOU	DICE / F1
DeepLabV3+ (ResNet-101)	Kleebauer (Fraunhofer)	175 MB	91.04%	0.9527
UNet++ (ResNet-34)	Moreno (UPM Madrid)	104 MB	88.50%	0.9390
YOLOv8-seg (ProtoNet)	Moreno (UPM Madrid)	6.7 MB	89.51%	0.9446
Global Renewables Watch	Microsoft AI for Good	180 MB	7.70%	0.0960
Baseline SegFormer (MiT-B0)	Moreno (UPM Madrid)	14.9 MB	87.20%	0.9316
Proposed Model (MiT-B2)	Our Work (Hugging Face)	94.4 MB	92.86%	0.9630

Note: Evaluated on the single frozen BDAPPV Google satellite test split (400 unseen tiles with positive arrays and hard negatives).

🔑 KEY DIFFERENCES IN OUR APPROACH

1. Continuous Multi-Scale Invariance

Prior papers trained on fixed flight altitudes. Our model trains across dynamic 0.4x to 1.6x scale pyramids, remaining invariant across Google Maps Zoom 18 to Zoom 20.

2. Authentic Hard-Negative Mining

Prior models trained almost exclusively on solar-positive images. We ingest ~800 real negative Google Earth tiles (empty roofs, skylights, roads), suppressing false-alarm spikes.

3. Boundary-Aware Compound Loss

Replaces standard Cross-Entropy with BCE + Soft Dice + Laplacian edge loss, producing sharp orthogonal GIS polygons ready for kilowatt capacity calculation.

Module 1: Geospatial Ingestion & Multi-Scale Pipeline

Purpose: Ingesting multi-source satellite streams, slicing high-res orthomosaics into 512×512 patches, and constructing scale-invariant training pyramids.

1. Multi-Sensor Data Stream Integration

Curates 3,000+ paired tiles combining BDAPPV Google Earth Engine (Google Maps satellite), PV01 (commercial flat concrete, corrugated metal, brick), and Davis CA aerial imagery.

2. Spatial Tiling & Overlapped Striding

Standardizes high-resolution imagery into 512×512 pixel patches with a 256px spatial stride (50% overlap) via Rasterio and GeoAI to prevent panel truncation along tile borders.

3. Scale-Invariant Pyramid Augmentation

Enforces dynamic random scaling (0.4x to 1.6x) and authentic hard negatives (~800 empty roofs, skylights, roads) to build resilience across Google Maps Zoom 18 to Zoom 20.

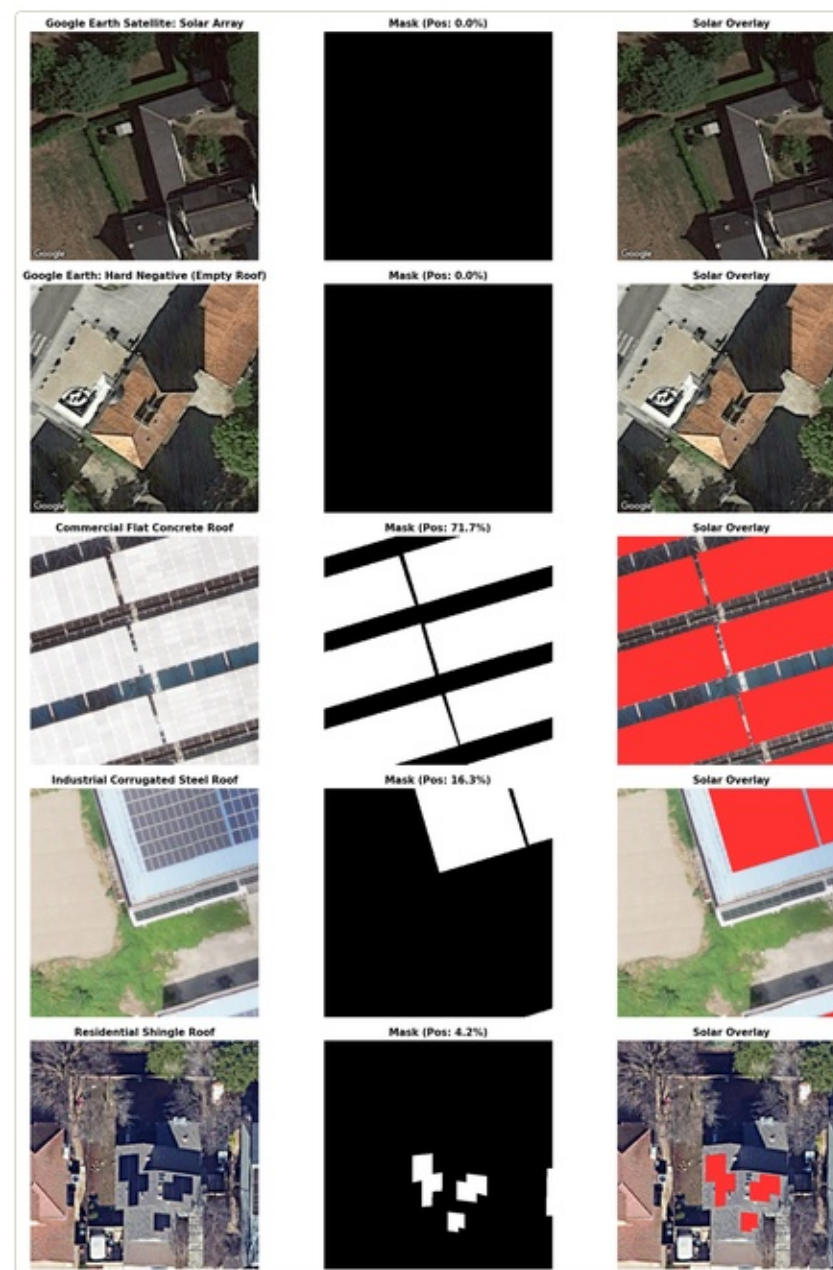


Fig: Training corpus diversity spanning Google Earth satellite arrays, hard negatives (empty roofs), commercial flat concrete, industrial steel, and residential shingle roofs.

Module 2: SegFormer Architecture & Boundary Loss

Purpose: Hierarchical self-attention feature modeling, multi-level context fusion, and boundary-aware edge regularization for sharp panel borders.

1. Hierarchical Mix Transformer (MiT-B2)

Replaces rigid convolutions with self-attention; captures global roof context without positional embedding limits across 4 hierarchical stages (1/4 to 1/32).

2. Lightweight All-MLP Decoder Head

Combines multi-level contextual features (~24.7M params) to maintain real-time throughput at ~29.6 FPS (33.8 ms/tile) on accessible Colab T4 GPUs.

3. Boundary-Aware Compound Loss

Formulates $0.4 \text{ BCE} + 0.4 \text{ Soft Dice} + 0.2 \text{ Laplacian edge loss}$. Penalizes boundary smearing and produces crisp, orthogonal building borders on rooftop arrays.

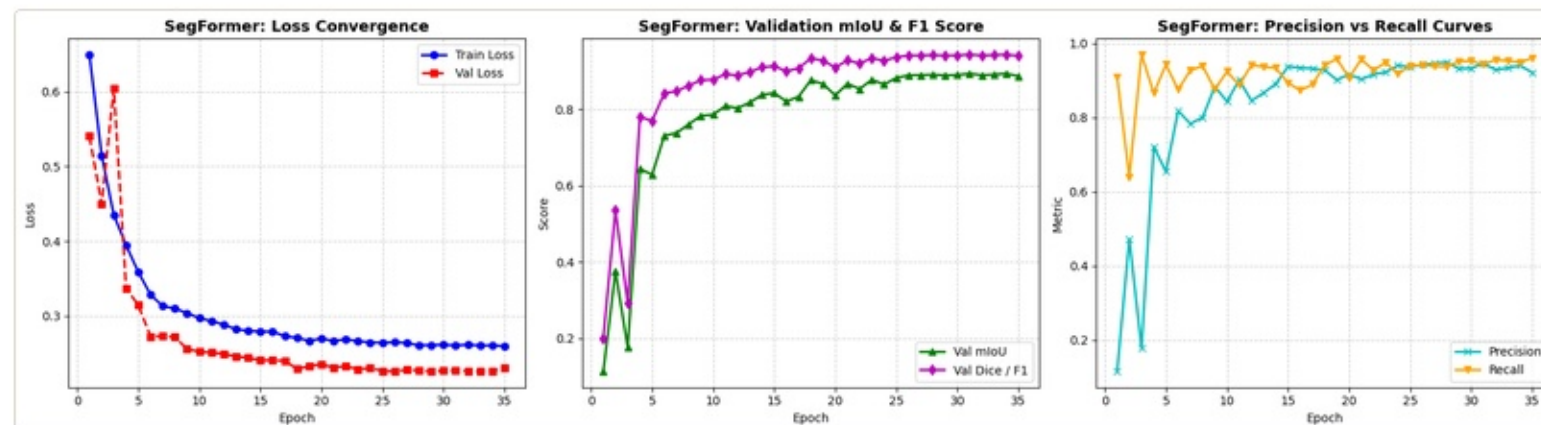


Fig: Research convergence curves – Loss convergence (left), Validation mIoU reaching 92.86% (center), and balanced Precision vs. Recall dynamics (right).

Module 3: Vectorization & Solar Capacity Engine

Purpose: Multi-scale test-time augmentation, converting raster masks into georeferenced GIS vector polygons (.geojson), and calculating municipal kilowatt capacity.

1. Multi-Scale Test-Time Augmentation (TTA)

Ensembles predictions across 0.75x, 1.0x, and 1.25x scale pyramids with morphological noise filtering to eliminate isolated 1-pixel false alarms on arbitrary Google Maps captures.

2. GIS Polygon Orthogonalization

Extracts contour vertices and regularizes organic raster outlines into clean right-angled polygons via GeoAI and Shapely, exporting georeferenced GeoJSON layers.

3. Solar Power Capacity Estimation Model

Integrates projected metric surface area (m^2) with local solar irradiance coefficients to calculate real-world generation capacity (kW / MW) and estimated annual yield (kWh/year).



Fig: Automated satellite detection comparison – Unprocessed raw satellite tile (left) vs. Delineated solar array instance masks (right).

References & Literature Citations

Key peer-reviewed journal publications, benchmark datasets, and technical documentation foundational to this research.

[1] Kasmi, G. et al. (2023)

A crowdsourced dataset of aerial images with annotated solar photovoltaic arrays and installation metadata.

Nature Scientific Data, 10(1), p.78. DOI: 10.1038/s41597-023-01951-4 (BDAPPV Benchmark).

[2] Jiang, H., Yao, L., & Liu, Y. (2021)

Multi-resolution dataset for photovoltaic panel segmentation from satellite and aerial imagery.

Earth System Science Data (ESSD), 13(11), pp.5389-5401. DOI: 10.5194/essd-13-5389-2021 (PV01).

[3] Yu, J. et al. (2018)

DeepSolar: A machine learning framework to analyze local solar deployment across the United States.

Joule (Cell Press), 2(12), pp.2605-2617. Stanford University Energy Institute.

[4] Wu, Q. (2026)

GeoAI: A Python package for integrating artificial intelligence with geospatial data analysis.

Journal of Open Source Software (JOSS), 11(118), p.9605 (Spatial Tiling & Orthogonalization Engine).

[5] Xie, E. et al. (2021)

SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers.

Advances in Neural Information Processing Systems (NeurIPS 2021), 34, pp.12077-12090.

[6] Kleebauer, M. et al. (2023)

Multi-Resolution Photovoltaic System Segmentation from Aerial and Satellite Imagery.

Fraunhofer Institute for Solar Energy Systems (ISE), Technical Report & Open-Source Benchmark.

[7] Moreno, J. (2023)

Deep Learning Framework for Automated Solar Mapping from Overhead Imagery.

ETSIT, Universidad Politécnica de Madrid (UPM), Master's Thesis & Open Model Weights.

[8] Microsoft AI for Good & Planet Labs (2023)

Global Renewables Watch: High-Resolution Planetary Mapping of Solar Energy Installations.

Nature / Microsoft AI for Good Open Source Release & Model Weights.

Thank You.

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